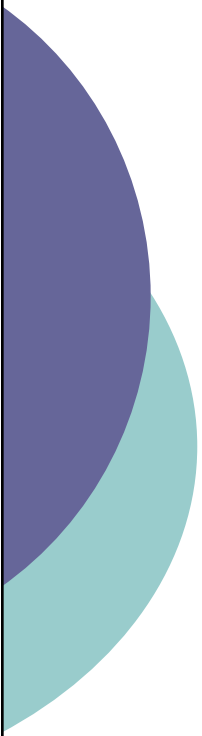


Bio 40C: Ovarian Physiology & Childbirth

Jeff Schinske



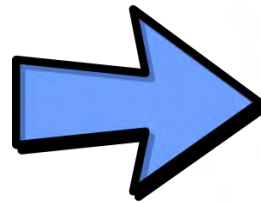


Bio 40C Proposed Norms

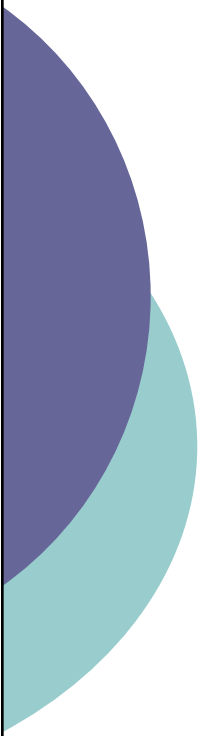
- Seek to enjoy your colleagues and your experiences in class!
- Actively select language that is inclusive, respectful, and welcoming
- Assume good intentions...no heroes and no villains (in this class)
- Recognize that we all have strengths as well as areas for growth
- Be curious, be skeptical
- Ask for help (and expect to receive it!)

Share with a partner...

1. Name
2. Something important to know about you as a person
3. What final exams, papers, or projects do you have coming up over the next week?

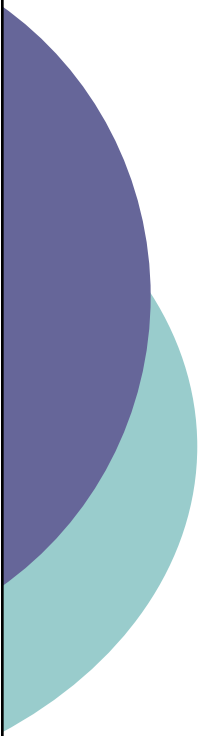


Person with 1st name closest to end of alphabet goes first



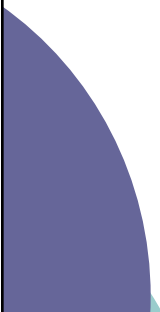
Debriefing Last Week

- Thanksgiving holiday + class cancelled due to illness = no new anonymous index cards or Canvas self evaluations to review!



A Brief Preview of the Week

- Today
 - Review of Application Question 13.
 - Considering ovulation and childbirth
- Lab This Week
 - Lab test
- Online Lecture Work This Week
 - End of course reflections
 - Revising application questions
- Next Week
 - Final exam meeting Tuesday at 1pm



Final “Exam” Activities

Application Questions 1-14 (60 pts)

In the afternoon/evening of December 12th, I will randomly select six of the fourteen application questions we have completed. Your scores on those application questions will, at that time, be counted toward the final exam. If you received 10/10 on your application questions (or 7/10 for a late application question), you will receive *full credit* (60/60) for this activity. If you do not yet have 10/10 or 7/10 on an application question, be sure to revise and resubmit those responses prior to 1pm on December 12th. Of course, if your revisions come in on the 12th, I will not have time to review them and provide feedback before 1pm. If you're concerned your revised answers will not get you to a mostly complete response, you might wish to submit your revisions earlier so there's time to receive feedback on those revisions and potentially resubmit them again.

In Person “Final” (60 points)

I want to take full advantage of this final opportunity to gather our community! Our on-campus final is scheduled for Tuesday, December 12th, from 1:00-3:00. We will meet in our usual lecture room (8402) at 1:00 on the 12th. At that time, everyone will participate in some collaborative activities related to our key Bio 40A goal - exercising our cellular/chemical A&P perspectives! You will receive full credit (60 points) for fully participating in these activities. In other words, credit will be awarded based on your thoughtful and full participation, not based on “correctness” of responses. *No advance preparation is required for this in-person part of the final.* Please simply join us that day!

I ask that on 12/12 you bring with you a fully charged, web-enabled device (e.g., smartphone, tablet, laptop), which you will use for web research. If this will pose a challenge for you, please let me know in advance so I can accommodate you!

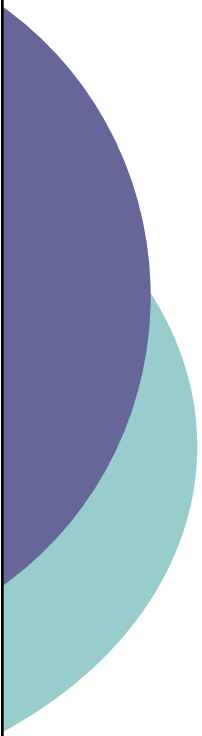
Application Question 13



Thinking as always through our "A&P perspective," what reproductive system issues do you think could arise from an injury like the one happening in the picture to the left?

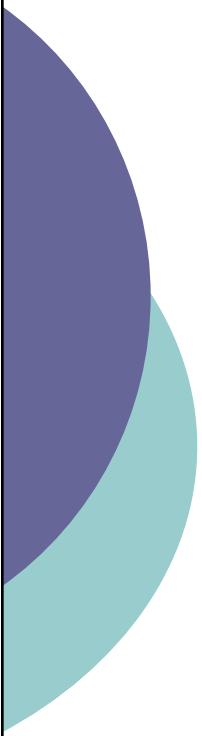
See if you can describe the involvement of Nurse Cells and white blood cells in this scenario.

"This injury could cause damage and inflammation to the blood testis barrier, which is normally established by the nurse cells. This will let white blood cells past the blood testis barrier, which could result in infertility."



Our Specific Outcomes

- Today you will be able to...
 - Justify the steps in *in vitro* fertilization through an A&P perspective on physiology in the ovary.
 - Describe the physiological rationale for the use of oxytocin during childbirth.



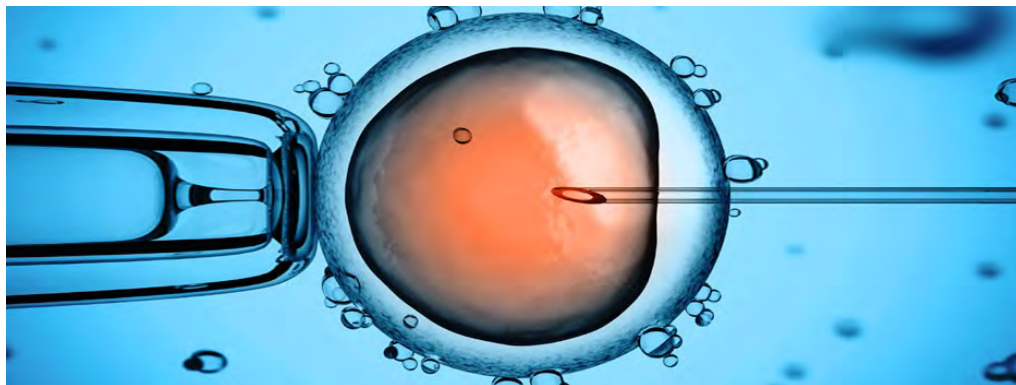
A Reminder of Our Disclaimer...

- “Male” will refer to the most common A&P of humans that produce sperm.
- “Female” will refer to the most common A&P of humans that produce eggs.
- But wide variation exists and humans do not fall into binary sexes.

The A&P perspective on *in vitro* fertilization

Adel is a 32 year old female excited about having a child. She has identified a sperm donor and has been preparing for *in vitro* fertilization. As part of this procedure, her eggs will be collected and fertilized in the lab. Then embryos will be placed in her uterus for development and birth.

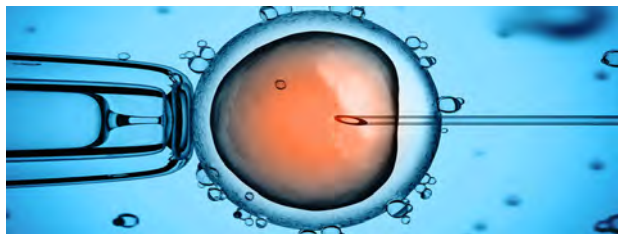
Adel has done lots of research and really wants to understand all the biology behind the procedure. We will try to help answer some of Adel's questions.



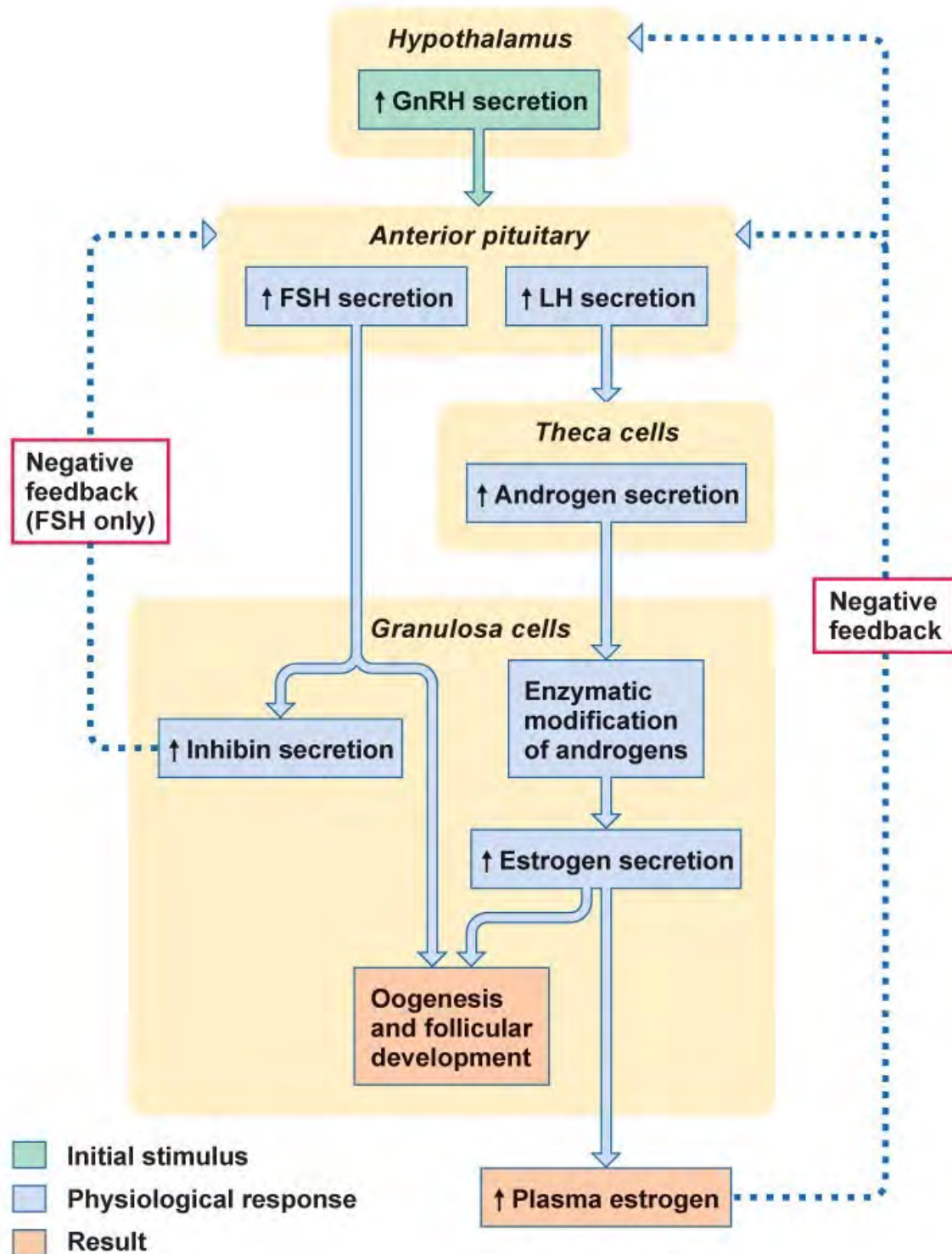
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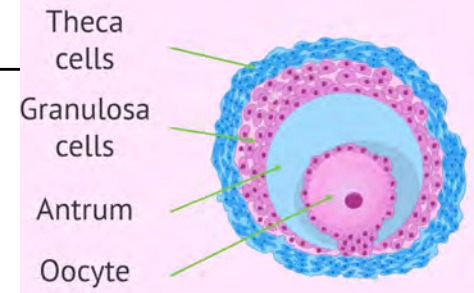
Adel has done lots of research and really wants to understand all the biology behind the procedure. We will try to help answer some of Adel's questions.



Adel says, "I read that I will soon have to go through something called 'Ovarian Reserve Testing' to see if I have plenty of eggs left in my ovaries. That makes sense to me. **It says that a good result would mean low FSH and high AMH? What are those hormones and why would I want one to be high and the other low?**"



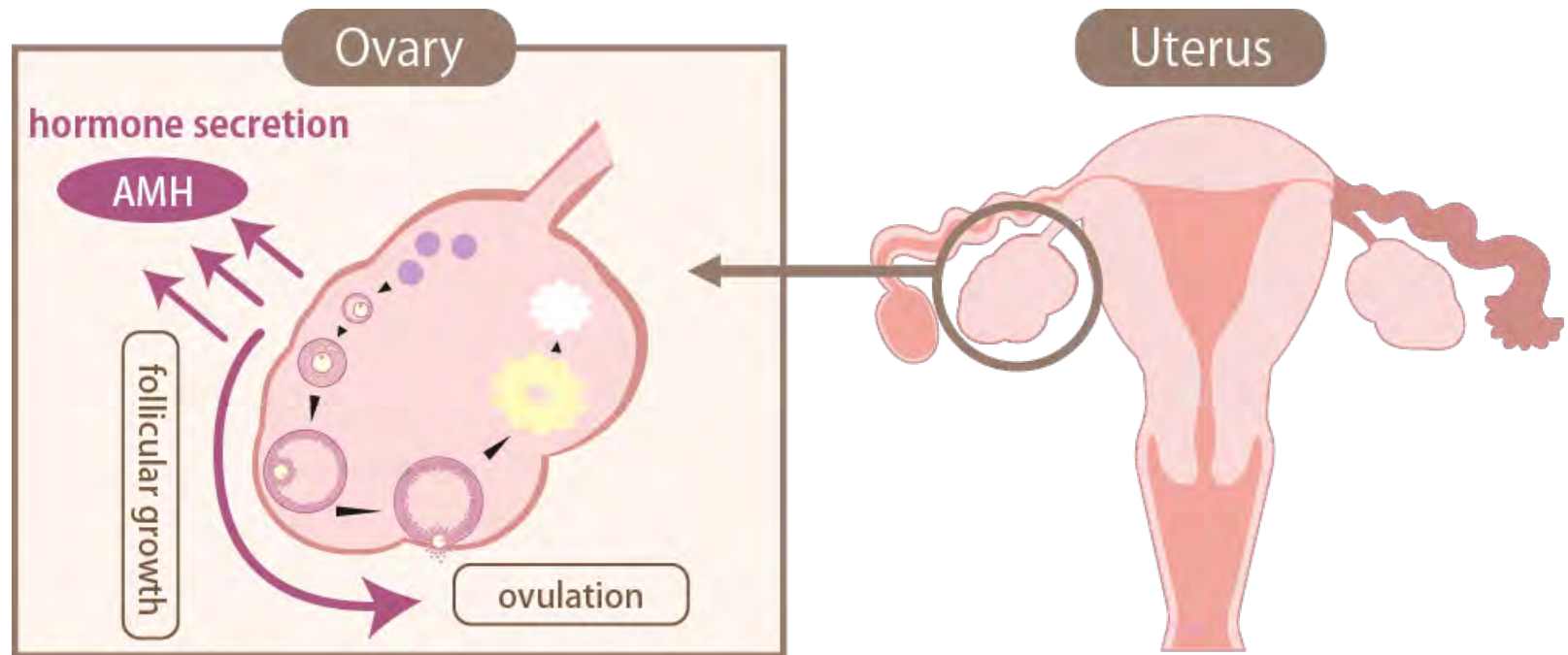
Reminder: A follicle includes an immature egg cell and supporting cells that produce hormones.



If Adel's pituitary is releasing lots and lots of FSH, that means the pituitary is **not being inhibited very much. Think about what would normally inhibit the pituitary.**

With that in mind, how could high FSH relate to a low number of eggs in the ovaries?

The A&P perspective on *in vitro* fertilization

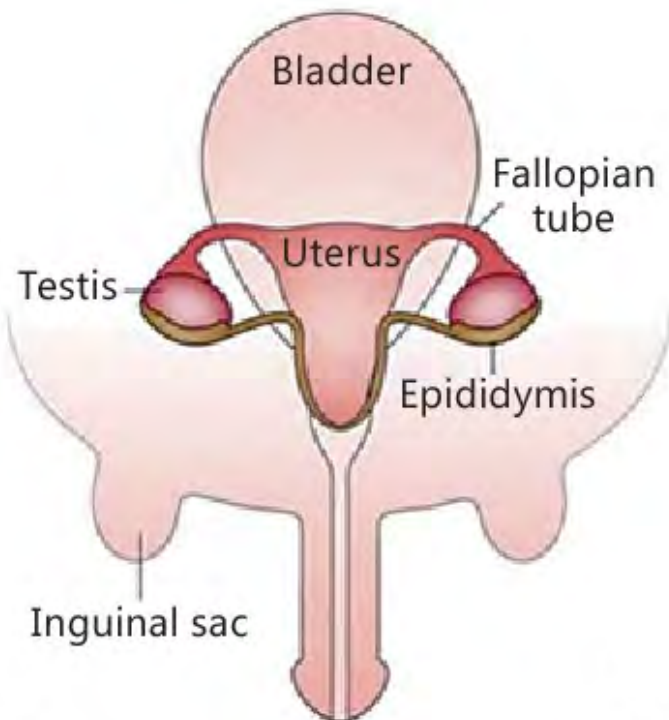


AMH is produced by early stage follicles, so more AMH = more follicles.

A Brief Aside Regarding AMH

In male fetuses AMH suppresses development of a uterus and uterine tubes.

If a fetus with XY chromosomes has an atypical version of AMH or less effective AMH receptors, the individual might be intersex.

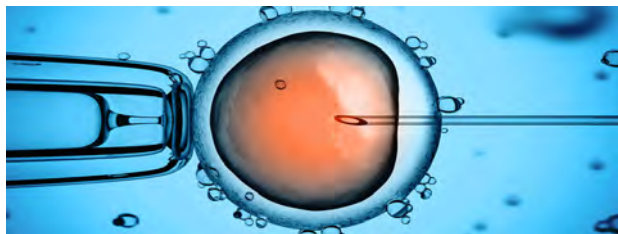


As we've reiterated in this class, biological sex is nonbinary and includes wide variation!

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Adel has done lots of research and really wants to understand all the biology behind the procedure. We will try to help answer some of Adel's questions.



Adel replies, "Thank you! That makes sense. If I have enough follicles in my ovaries, I understand I will then have 'Ovarian Stimulation' where I get extra FSH to create more and more mature follicles."

"Won't ovarian stimulation reduce my reserve of eggs and make me run out of eggs faster?"

Considering the Ovaries...

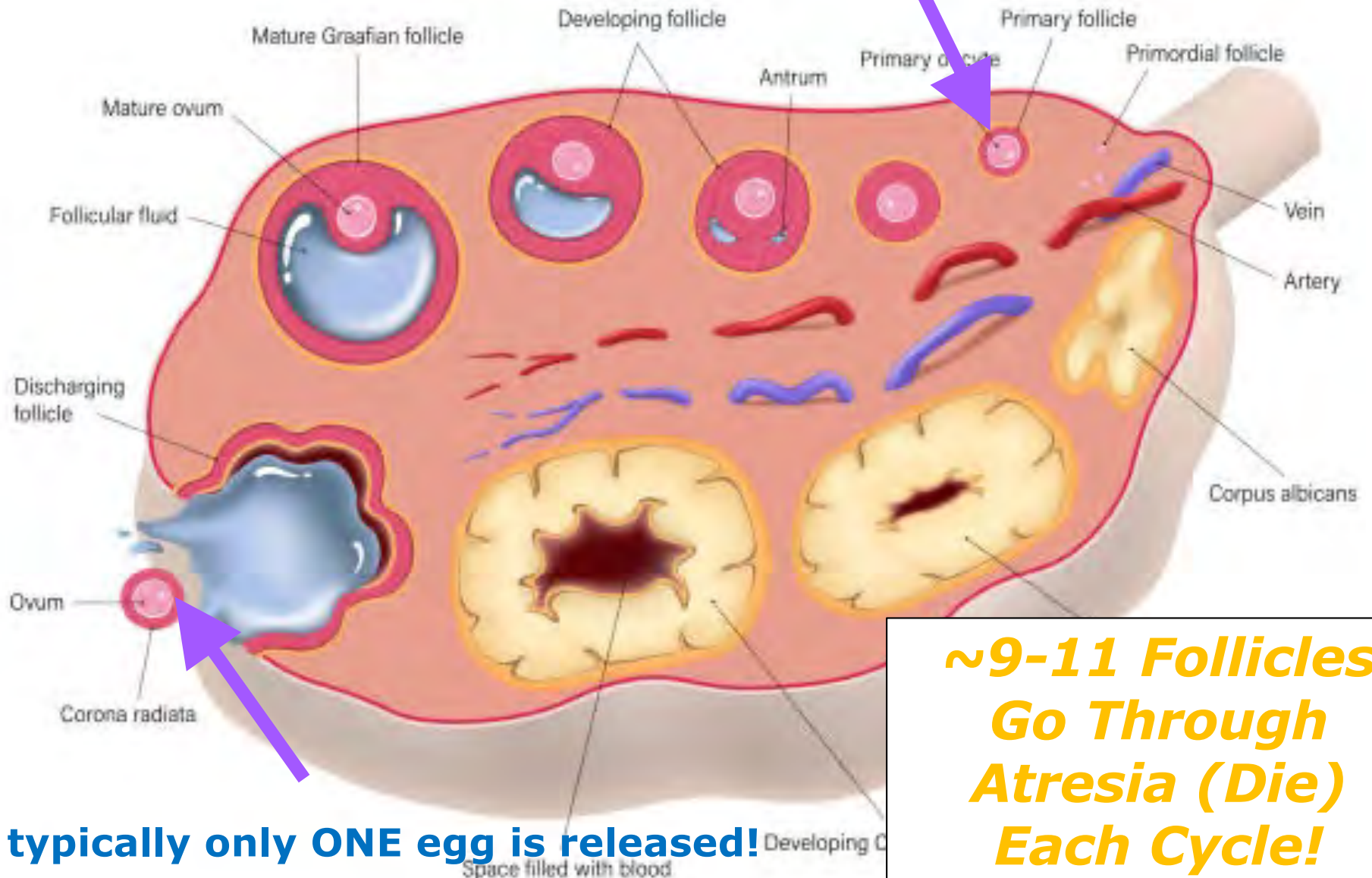
At Birth: Millions of Primordial Follicles

Start of Puberty: ~500,000 Primordial Follicles

At Menopause: No Primordial Follicles

Considering the Ovaries

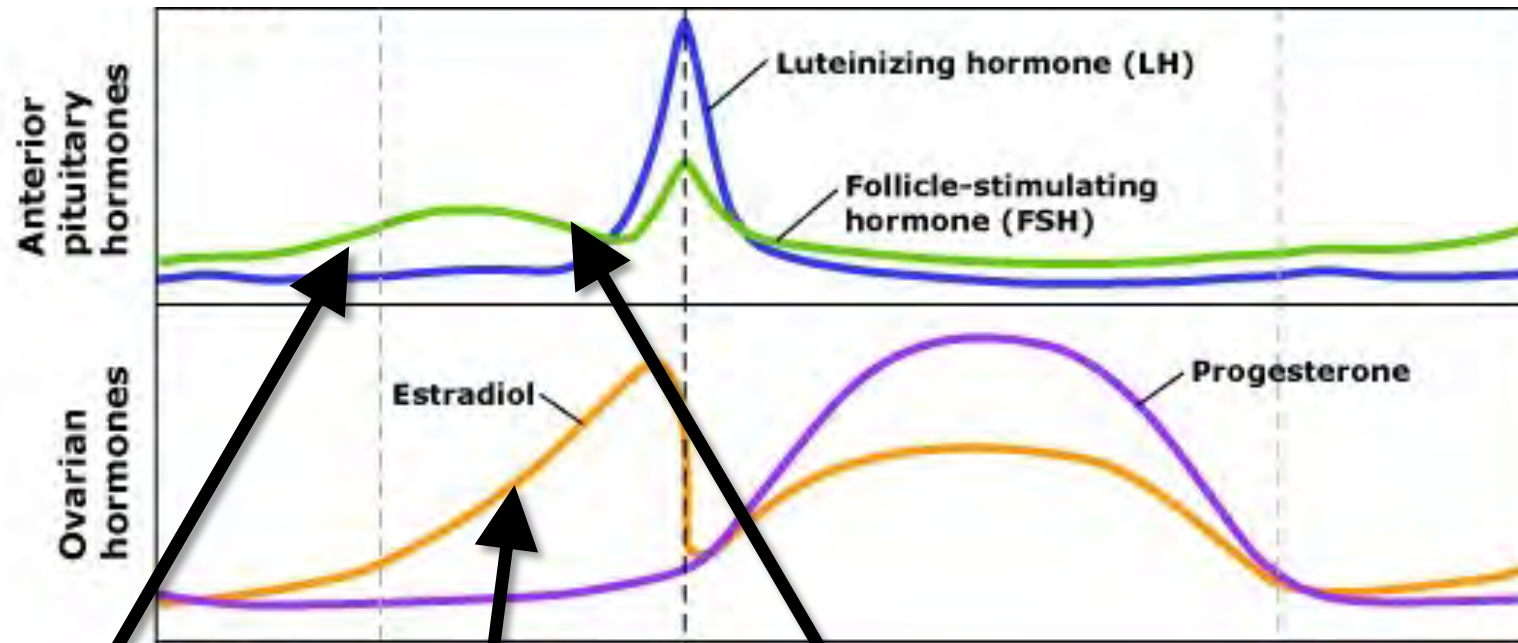
Each cycle 10-12 follicles begin growing



But typically only ONE egg is released!

**~9-11 Follicles
Go Through
Atresia (Die)
Each Cycle!**

FSH & Atresia



FSH stimulates follicles to development

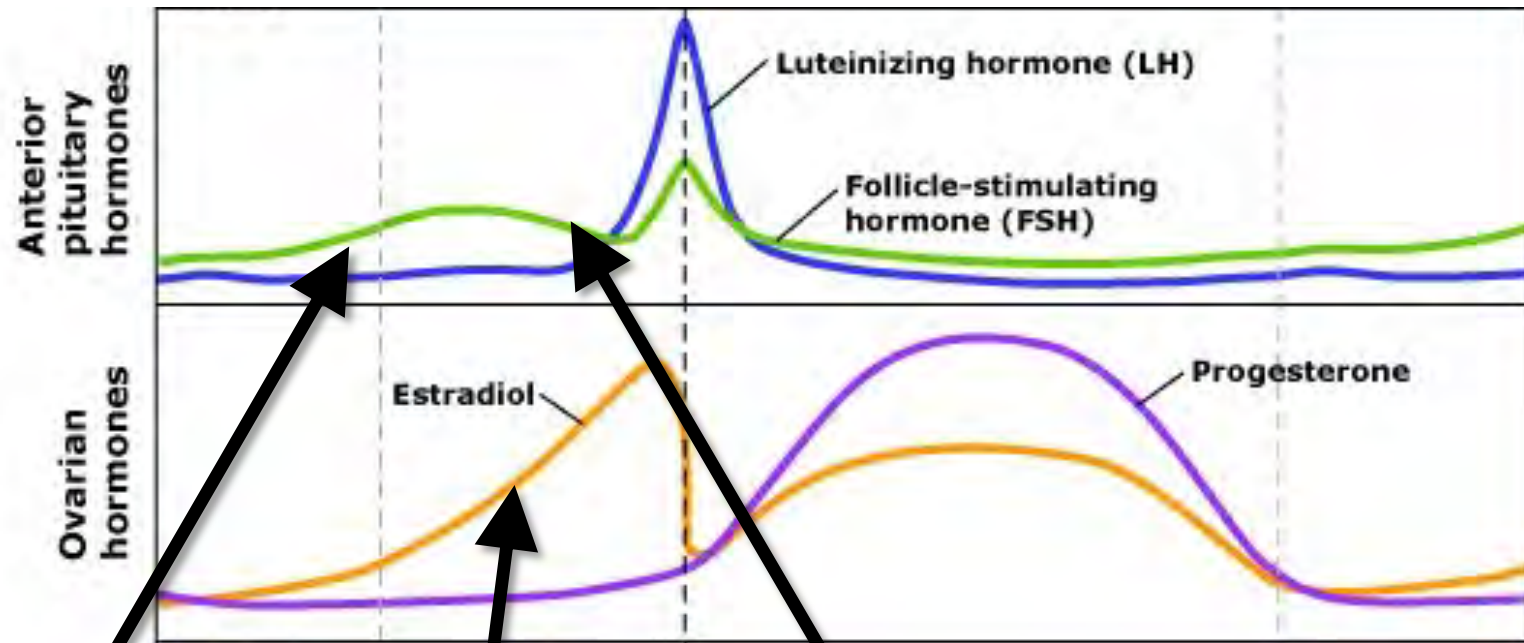
Developing follicles release estrogen

Estrogen causes FSH to decrease

Follicles with fewer FSH receptors are starved of FSH stimulation and go through atresia

Surviving follicle has most FSH receptors.

So what would happen if the ovaries received unlimited FSH?



FSH stimulates follicles to development

Developing follicles release estrogen

What if there was no decrease here?

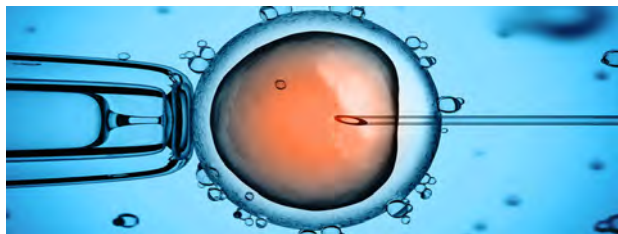
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Surviving follicle has most FSH receptors.

The A&P perspective on *in vitro* fertilization

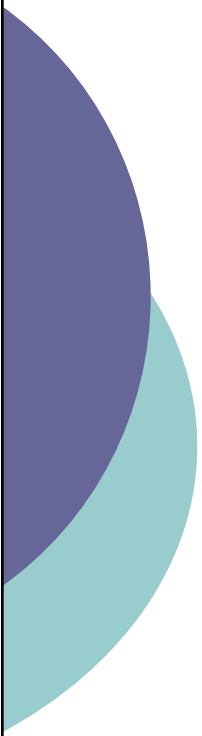
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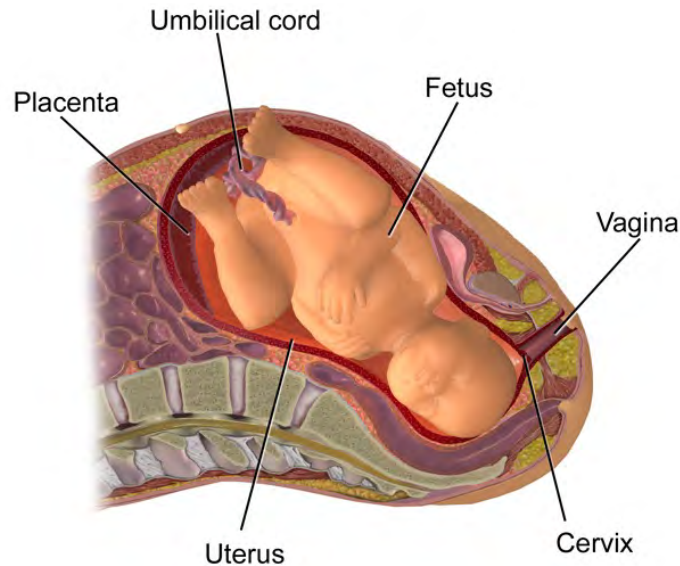
The A&P perspective on *in vitro* fertilization

Let's say Adel successfully became pregnant and is child is nearly ready to be born! Yay!

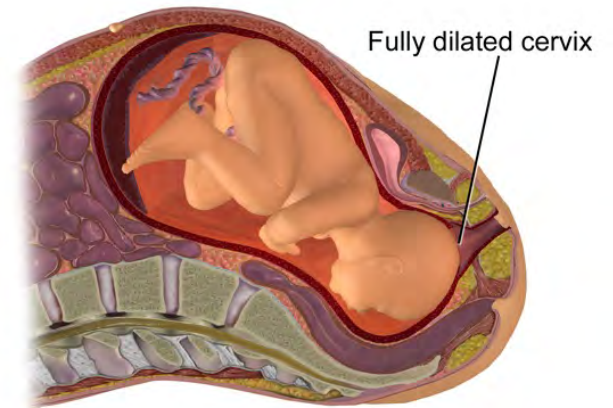
Adel's doctors told her she might receive oxytocin during childbirth and Adel wants to know why.

Changes in the levels of estrogen vs. progesterone make the uterus more sensitive to contracting

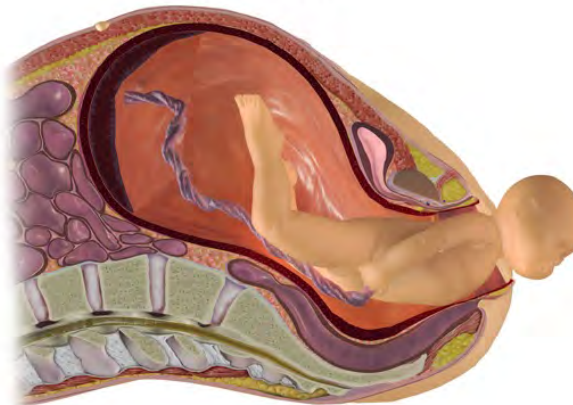
Resulting in Childbirth!



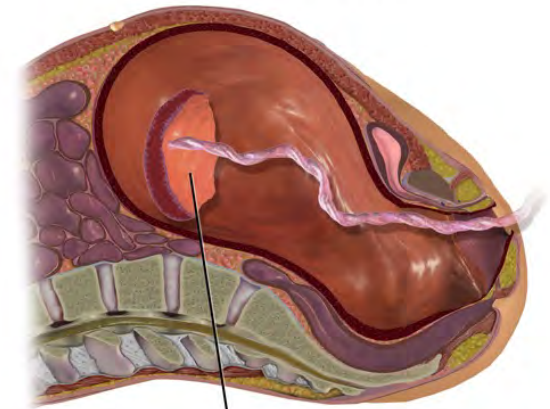
Stage 1: Dilation



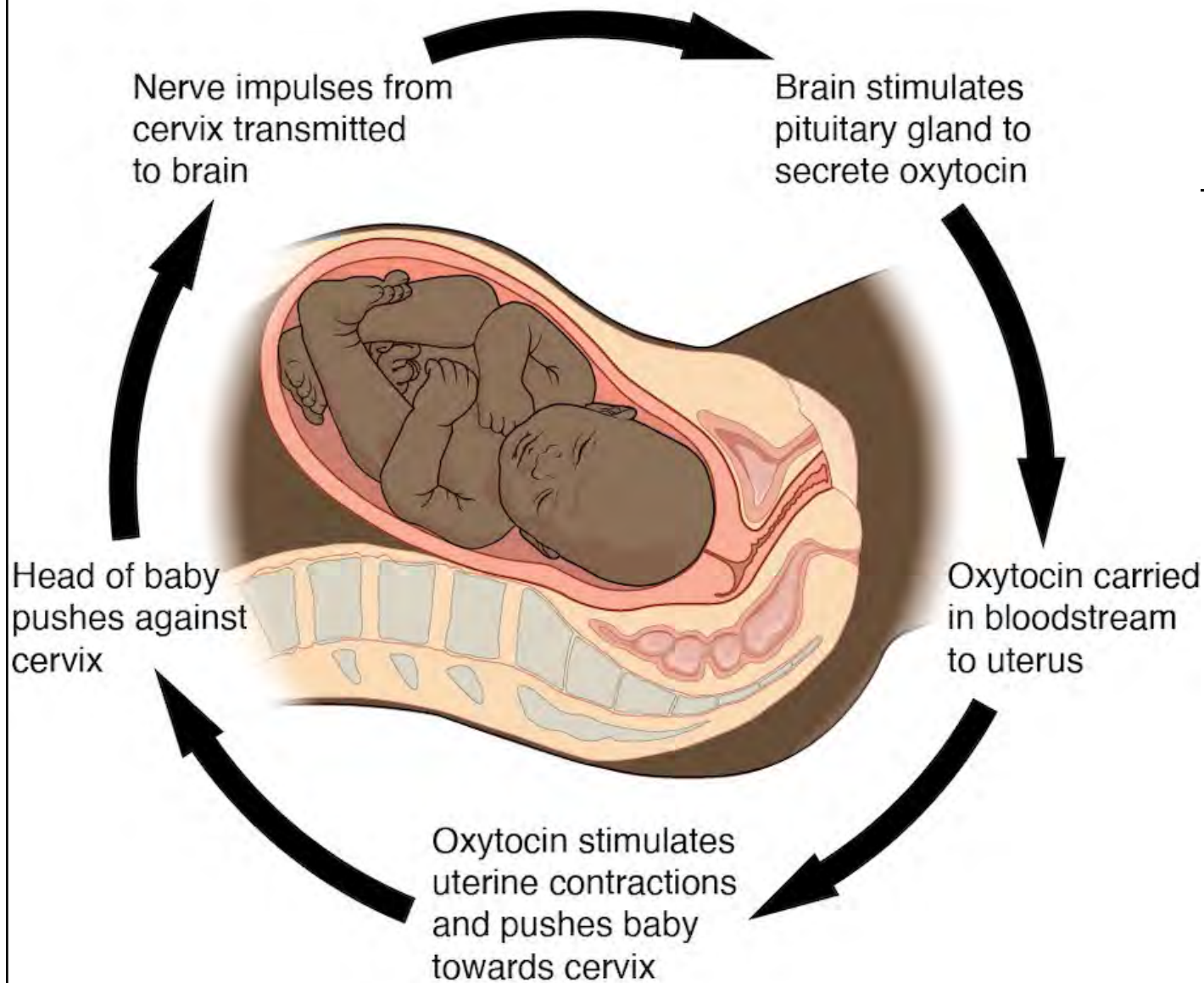
Stage 2: Birth



Stage 3: Afterbirth

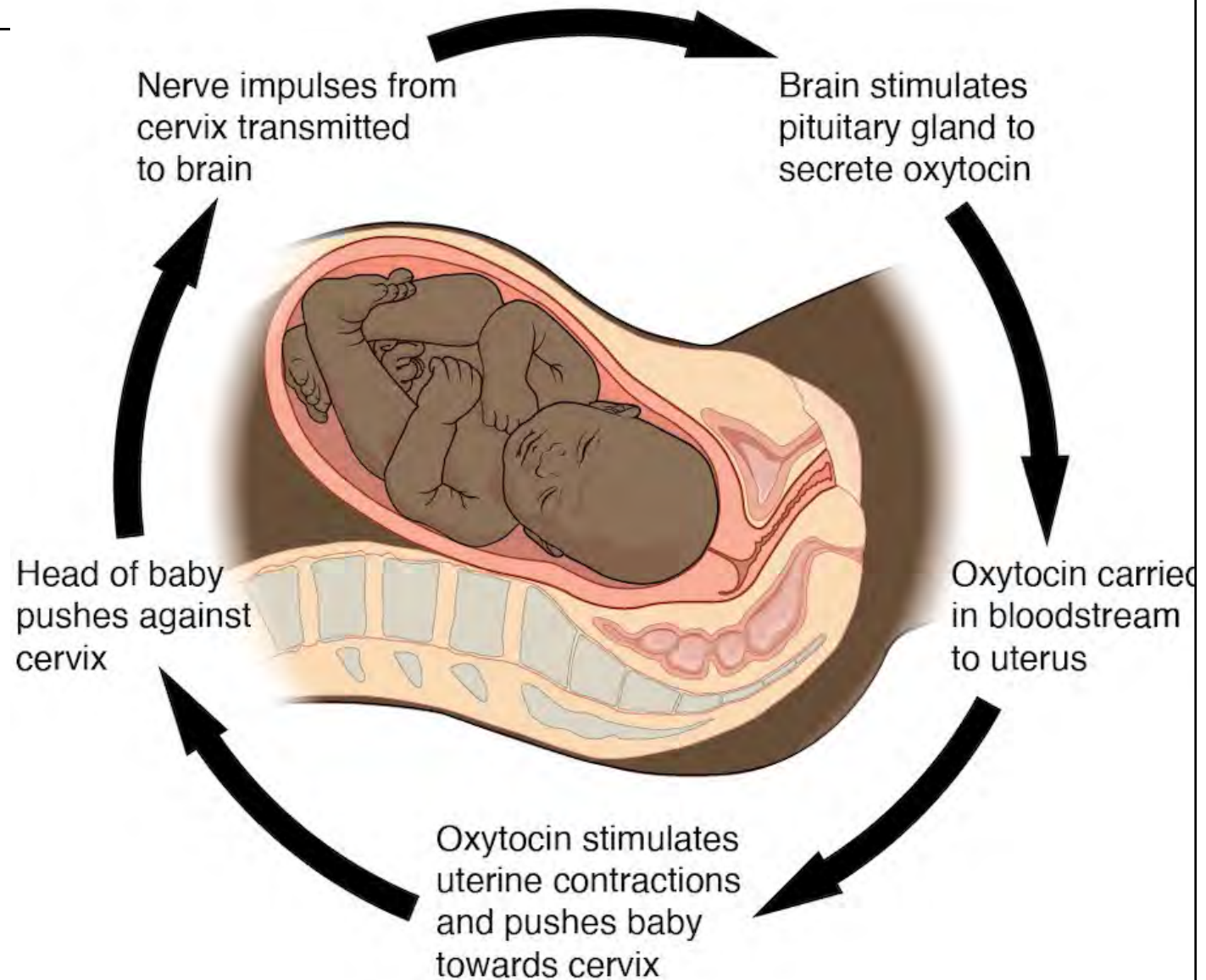


Placenta detaches and exits vagina

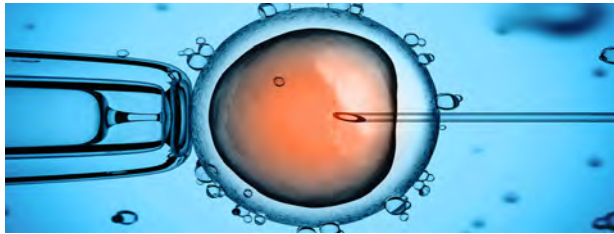


Positive Feedback Loop During Labor

Why might doctors administer oxytocin during labor?



The A&P perspective on *in vitro* fertilization



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"Won't ovarian stimulation reduce my reserve of eggs and make me run out of eggs faster?"

How would you respond to Adel?

14 **Your Name**

Your responses...

Use the terms

- **Primordial follicles**
- **FSH**
- **Atresia**